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Externalizing behavior, the UPPS-P Impulsive Behavior scale and Reward and Punishment Sensitivity

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ABSTRACT

The UPPS-P Impulsive Behavior scale is a promising measure of impulsivity facets involved in the etiology of Externalizing behaviors (e.g., substance abuse, antisocial behavior, and aggression). The purpose of this study was to determine whether measures of motivational systems, specifically Reward and Punishment Sensitivity, accounted for variance in Externalizing behaviors beyond the UPPS-P scales. Undergraduates ($N = 282$; 50% men) completed online self-reports of alcohol and drug use, antisocial behavior, and aggression, as well as the UPPS-P and measures of Reward and Punishment Sensitivity. Principal components analysis was used to create summary Externalizing scores related to Disinhibition and Aggression. Lack of Premeditation, low Perseverance and Sensation Seeking had significant partial relationships with Disinhibition, but Reward Sensitivity accounted for additional variance. Both Reward Sensitivity and (Low) Punishment Sensitivity were related to Aggression, beyond variance explained by UPPS-P Negative Urgency and (lack of) Perseverance. Impulsivity facets appear to have differential relationships with measures of Externalizing and, although the UPPS-P accounts for a significant portion of individual differences, it does not fully account for variance associated with Reward and Punishment Sensitivity.

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1. Introduction

Externalizing behaviors, such as delinquency, aggression and substance use, are related to impulsive traits (e.g., Krueger, Markon, Patrick, Benning, & Kramer, 2007). However, impulsivity has been conceptualized in many ways and appears to be heterogeneous (Miller, Flory, Lynam, & Leukefeld, 2003). Whiteside and Lynam (2001) developed the UPPS Impulsive Behavior scale to identify and assess dissociable impulsive traits either not adequately captured with, or delimitated by, existing scales. A multifaceted, comprehensive measure of impulsivity should improve our understanding of which aspects are most relevant in the etiology of Externalizing behavior (Krueger et al., 2007). The UPPS was a major step in this direction as it summarized traits captured by a set of well-established measures while increasing the possibility for discriminant validity by minimizing correlation amongst its factor analytically-derived scales.

Several theories of antisocial personality, however, posit etiological roles for motivational systems derived from the Reinforcement Sensitivity Theory (RST; Gray & McNaughton, 2000; see Fowles, 1988; Lykken, 1995). In the RST, the Behavioral Activation System (BAS) is thought to be sensitive to cues of reward or negative reinforcement. When activated, individuals either engage in approach related behavior towards the likely reward or actively avoid a potential punisher. The Behavioral Inhibition System (BIS) was traditionally viewed as being sensitive to classically conditioned cues of punishment or frustrative non-reward, but is now viewed as playing an inhibitory role when conflicts arise between BAS activation and activation of a third system, called the Fight-Flight-Freezing System (FFFS), which is sensitive to aversive stimuli and cues thereof. Individual differences exist in typical tonic levels of these systems. Traditional theoretical adaptations of the RST used to explain Externalizing behavior have focused on the BAS's sensitivity to reward and the BIS's sensitivity to classically conditioned cues of punishment. In the contemporary view, punishment sensitivity may reflect combined functioning of the FFFS and the BIS (Corr, 2004). Factor analyses of impulsivity scales including measures of these systems identify dissociable dimensions related to Reward Sensitivity, Punishment Insensitivity, and a kind of Rash Impulsivity not covered by the RST, related to acting without considering the consequences (e.g., Caseras, Avila, & Torrubia, 2003; Franken & Muris, 2006). The scales that went into the development of the UPPS, however, arguably underrepresented

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Punishment and Reward Sensitivity. It is unclear if putative measures of Reward and Punishment Sensitivity account for variance in Externalizing behaviors not captured by the UPPS.

In developing the UPPS, Whiteside and Lynam (2001) looked at impulsivity in terms of the Five Factor Model and major measures of impulsivity. They used exploratory factor analysis, which uncovered four factors. First, Premeditation is the tendency to think ahead and plan carefully. Second, Urgency, now called Negative Urgency, reflects a tendency to commit rash actions when experiencing negative affect. Third, Sensation Seeking is a tendency to seek out stimulation or excitement. Fourth, Perseverance is the ability to attend to a task without getting bored. Cyders et al. (2007) subsequently added a facet, Positive Urgency, indexing a tendency to act rashly when experiencing positive affect.

Lack of Premeditation and high Sensation Seeking have been related to alcohol use and abuse in young adults (Lynam & Miller, 2004; Magid & Colder, 2007). Lack of Premeditation, Negative Urgency and Sensation Seeking were related to drug use, and lack of Premeditation was related to marijuana abuse in adults (Lynam & Miller, 2004). Although all facets may be zero-order correlates of substance use in college students, Negative and Positive Urgency predict increases in drinking (Settles, Cyders, & Smith, 2010) and drug use across a school year (Zapolski, Cyders, & Smith, 2009). Aggression has been related positively to Negative Urgency and antisocial behavior is linked to Sensation Seeking and a lack of Premeditation (Lynam & Miller, 2004; Miller et al., 2003). All of the original UPPS scales distinguished antisocial alcohol abusers from non-antisocial alcohol abusers (Whiteside & Lynam, 2003).

High Reward Sensitivity has been related to substance use and disorder, aggression and antisocial behavior (Bjørnebekk, 2007; Franken, Muris, & Georgieva, 2006; Hasking, Scheier, & Ben Addallah, 2011). Although, not universally supported, some studies reporting Reward Sensitivity correlates of Externalizing also report associations with low Punishment Sensitivity (e.g., Franken & Muris, 2006; Taylor, Reeves, James, & Bobadilla, 2006; Voigt et al., 2009). Some studies, however, suggest that Punishment Sensitivity may not be uniquely associated with Externalizing once statistically controlling for Reward Sensitivity (e.g., Hundt, Kimbrel, Mitchell, & Nelson-Gray, 2008).

In the present study, we administered the revised UPPS, the UPPS-P, measures of Reward and Punishment Sensitivity, and self-report of substance use, aggression, and antisocial behavior to a sample of undergraduates. We predicted significant zero-order correlations between an Externalizing dimension and all five UPPS-P scales as well as Reward and Punishment Sensitivity. Given consistent past partial associations of low Premeditation and high Sensation Seeking with a multitude of Externalizing behaviors, we predict that these scales will have significant unique associations with Externalizing. Given the importance of Reward Sensitivity to theories of Externalizing and Reward Sensitivity emerging as a separate trait from facets like those in UPPS-P, we predict that Reward Sensitivity will be related to variance in Externalizing not accounted for by the UPPS-P.

2. Method

2.1. Participants and procedure

Two hundred eighty-two undergraduate students (50% male) participated in this study. They were between the ages of 19 and 30 (M = 20.79 years, SD = 2.11). Students were required to speak English as their primary language for at least 10 years. The majority self-identified as being of European (43.3%), or East Asian descent (40.1%). Students at a western Canadian university were recruited through a departmental website and participated for extra credit in a psychology class. Procedures were approved by an institutional research ethics board. A participants provided informed consent and completed self-reports online using the website http://www.SurveyMonkey.com.

2.2. Measures

2.2.1. UPPS-P Impulsive Behavior scale

The UPPS-P included the 45-items of Whiteside and Lynam (2001) combined with the 14-item Positive Urgency scale of Cyders et al. (2007). The items were randomly ordered with the following Cronbach's Alphas: Premeditation (.88), Negative Urgency (.88), Sensation Seeking (.89), Perseverance (.85), and Positive Urgency (.93). In order to make scores sensible given the scale names we scored the scales so that high Negative Urgency, Sensation Seeking, and Positive Urgency indicated high impulsivity, whereas high Premeditation and Perseverance indicated low impulsivity.

2.2.2. Behavioral Inhibition System and Behavioral Activation System scales

The Behavioral Inhibition System and Behavioral Activation System Scales (BIS/BAS; Carver & White, 1994) are comprised of 20-items. The BIS scale (alpha = .80) is unidimensional. The BAS scale, however, consists of three subscales: Drive (alpha = .81), Fun Seeking (alpha = .80) and Reward Responsiveness (alpha = .82). Scales were scored so that high scores indicate high Punishment or high Reward Sensitivity.

2.2.3. Sensitivity to Punishment and Sensitivity to Reward Questionnaire

The Sensitivity to Punishment and Sensitivity to Reward Questionnaire (SPSRQ; Torrubia, Ávila, Moltó, & Caseras, 2001) is a 48-item measure. The Sensitivity to Punishment (SP, 24-items) and Sensitivity to Reward (SR, 24-items) scales were internally consistent (alphas = .85 and .80, respectively).

2.2.4. Aggression Questionnaire

The 9-item Physical Aggression (alpha = .86) scale and 5-item Verbal Aggression (alpha = .74) scale from the Aggression Questionnaire (Buss & Perry, 1992) were used to measure overt aggressive behaviors.

2.2.5. Self Reported Delinquency scale

The Self Reported Delinquency Scale (SDS; Elliott & Ageton, 1980) is a 47-item measure that taps antisocial behavior. The last 7 drug use items were excluded as this was measured elsewhere. Participants were required to indicate the frequency they engaged in each behavior in the past 5 years in an open-field format. Scores were the sum of the 40-items.

2.2.6. Substance use

Questions recommended by the US National Institute on Alcohol Abuse and Alcoholism (National Institute on Alcohol Abuse, 2003) were used to assess typical quantity of alcohol consumed per occasion and typical frequency of alcohol consumption in the past year. A quantity $\times$ frequency estimate of past-year use was created by multiplying quantity and frequency. An 11-item drug use frequency scale adapted from the Adolescent Drug Involvement Scale (ADIS; Moberg & Hahn, 1991), which will be referred to as the Drug Use Frequency (DUF) measure, was included. The question inquired about the following drugs: marijuana or hashish; LSD, psilocybin, peyote, other hallucinogens (acid); amphetamines (including meth); cocaine; barbiturates; PCP; heroin; other opiates; valium and other tranquilizers; ecstasy and inhalants (glue, gasoline, spray paint, etc.).

3. Results

3.1. Data reduction

We summarized the common variance in (1) measures of Externalizing behaviors and (2) Punishment and Reward Sensitivity using two Principal Components Analyses (PCAs) with Varimax rotation conducted with SPSS version 19 and saved component scores with the regression method (see Table 1). Most Externalizing variables were positively skewed. A natural log transformation was used for all except the Aggression scores. Two components from the PCA involving Externalizing measures had eigenvalues >1 and these accounted for 74.47% of the variance. Based on the pattern of loadings, these components will be referred to as Disinhibition and Aggression.

The second PCA produced two components with eigenvalues >1 and these accounted for 63.98% of the variance across BIS/BAS and SPSRQ scales. Based on the pattern of loadings, the components will be referred to as Reward and Punishment Sensitivity.

3.2. Zero-order correlations

Table 2 provides descriptive statistics and zero-order correlations. Disinhibition and Aggression were significantly correlated with all UPPS-P scales. Both Externalizing components were significantly positively correlated with Reward Sensitivity and negatively correlated with Punishment Sensitivity.

3.3. Partial relationships between Impulsivity and Disinhibition

A hierarchical multiple regression examined the partial relationships between Reward Sensitivity, Punishment Sensitivity, UPPS-P scales and Disinhibition (see Table 3). Sex and age were included in Step 1. UPPS-P subscales were added on Step 2, whereas Reward Sensitivity and Punishment Sensitivity were added in Step 3. The model at Step 1 did not account for a statistically significant amount of variance. At Step 2, UPPS-P scales significantly accounted for variance in Disinhibition, $\Delta R^2 = .21$, $F(5, 274) = 15.05$, $p < 001$. Premeditation was significantly negatively, and Sensation Seeking was significantly positively, related to Disinhibition. Addition of Punishment and Reward Sensitivity at Step 3 increased the variance accounted for by a small, but significant degree, $\Delta R^2 = .03$, $F(2, 272) = 5.12$, $p = .007$. In this step, Reward Sensitivity, lack of Premeditation and Sensation Seeking were significantly associated with higher levels of Disinhibition. Lack of Perseverance became associated with Disinhibition. Notably, with the UPPS-P subscales included, Punishment Sensitivity, a zero-order correlate, was no longer significantly related to Disinhibition. Overall, the final model accounted for a quarter of the variance in Disinhibition ($R^2 = .25$).

3.4. Partial relationships between Impulsivity and Aggression

Parallel analyses were conducted with Aggression. Results are presented in Table 3. A significant amount of variance in Aggression was accounted for at Step 1, $R^2 = .11$, $F(2, 279) = 16.59$, $p < .001$. Male sex was significantly associated with Aggression. The addition of the UPPS-P scales at Step 2 accounted for additional variance, $\Delta R^2 = .18$, $F(5, 274) = 13.77$, $p < .001$. Sex, Negative Urgency and Sensation Seeking were significantly positively associated with Aggression. At Step 3, addition of Punishment and Reward Sensitivity led to a significant, albeit small, increase in variance accounted for, $\Delta R^2 = .03$, $F(2, 272) = 5.92$, $p = .003$. Intriguingly, Sensation Seeking no longer had a significant partial relationship with Aggression. Further, Perseverance, which had not been significantly associated with unique variance in Aggression, now had a significant negative regression coefficient. The final model accounted for about a third of the variance in Aggression ($R^2 = .32$).

4. Discussion

We found two dimensions underlying Externalizing behavior. The first was more general, while the other was specific to aggression. Although our initial predictions focused on one Externalizing dimension, our finding of a distinction between aggressive and other forms of disinhibited behavior is in keeping with previous

Table 1
Principal-components analyses for Externalizing behaviors and Reward and Punishment Sensitivity Scales.

Variable	*M*	*SD*	Disinhibition loadings	Aggression loadings
Externalizing Behaviors				
Physical Aggression	20.94	7.46	0.13	0.87
Verbal Aggression	13.89	3.84	0.09	0.88
Alcohol Quantity × frequency[a]	4.29	2.21	0.86	0.10
Drug Use Frequency[a]	0.66	0.63	0.85	0.05
Self-Reported Delinquency Scale[a]	3.11	1.77	0.82	0.18
Eigenvalue			2.41	1.32
% of variance			48.12	26.35
			Reward Sensitivity Loadings	Punishment Sensitivity Loadings
Reward and Punishment Sensitivity				
BIS/BAS				
BIS	20.82	3.68	0.10	0.91
BAS-RR	17.21	2.36	0.80	0.31
BAS-D	11.41	2.18	0.75	−0.10
BAS-FS	11.97	2.48	0.75	−0.26
SPSRQ				
SP	11.96	5.57	−0.35	0.76
SR	12.66	4.61	0.59	−0.15
Eigenvalue			2.24	1.60
% of variance			37.27	26.71

Note: N = 282. BAS-RR = BAS Reward Responsiveness scale; BAS-D = BAS Drive scale; BAS-FS = BAS Fun Seeking; SP = Sensitivity to Punishment; SR = Sensitivity to Reward.
[a] Log-transformation applied prior to PCA (descriptive statistics on raw scores).

Table 2
Means, standard deviations and Pearson correlations involving Impulsivity, Disinhibition, and Aggression.

	Negative Urgency(NU)	Premeditation (Pre)	Perseverance (Pers)	Sensation Seeking (SS)	Positive Urgency (PU)	Reward Sensitivity (Rew)	Punishment Sensitivity (Pun)	Disinhibition (Dis)	Aggression (Agg)
NU	28.23 (6.73)								
Pre	−.40**	32.12 (5.46)							
Pers	−.47**	.43**	28.77 (5.25)						
SS	.16**	−.27**	.11	34.60 (7.19)					
PU	.73**	−.43**	−.38**	.28**	28.63 (8.58)				
Rew	.16**	−.12*	.21**	.47**	.15*	0 (1.00)			
Pun	.11	.36**	−.04	−.49**	−.09	.00	0 (1.00)		
Dis	.20**	−.31**	−.12*	.41**	.19**	.32**	−.22**	0 (1.00)	
Agg	.37**	−.17**	−.18**	.26**	.32**	.17**	−.20**	.00	0 (1.00)

Note: $n = 282$. Means and standard deviations are provided on the diagonal.
* $p < .05$.
** $p < .01$.

Table 3
Hierarchical multiple regressions predicting Externalizing behavior from Impulsive traits.

	Disinhibition		Aggression	
	ΔR^2	β	ΔR^2	β
Externalizing behavior				
Step 1	.02		.11***	
Sex		−.12		−.33***
Age		.06		−.01
Step 2	.21***		.18***	
Sex		−.09		−.34***
Age		.04		−.01
Negative Urgency		.11		.36***
Premeditation		−.18**		.00
Perseverance		−.08		−.09
Sensation Seeking		.36***		.14*
Positive Urgency		−.10		−.04
Step 3	.03**		.03**	
Sex		−.11		−.33***
Age		.05		−.02
Negative Urgency		.07		.41***
Premeditation		−.16*		.08
Perseverance		−.14*		−.13*
Sensation Seeking		.27***		.00
Positive Urgency		−.09		−.05
Reward Sensitivity		.20**		.15*
Punishment Sensitivity		−.03		−.21**

Note: $n = 282$. Regression coefficients are standardized.
* $p < .05$.
** $p < .01$.
*** $p < .001$.

conceptualizations (e.g., Tackett, Krueger, Iacono, & McGue, 2005). As expected, significant correlations were found between both Externalizing components and all impulsivity-relevant traits. Consistent with past partial associations, low Premeditation and high Sensation Seeking were significantly associated with Disinhibition and Negative Urgency was associated with Aggression.

In line with the prediction that Reward Sensitivity would account for unique variance in Externalizing beyond that of the UPPS-P scales, high Reward Sensitivity was related to Aggression and Disinhibition. Low Punishment Sensitivity was also significantly associated with Aggression. The UPPS-P did not account for all of the covariance between Reward Sensitivity and either form of Externalizing behavior.

4.1. Disinhibition

Consistent with past findings, Disinhibition was associated with high Sensation Seeking and lack of Premeditation (Lynam & Miller, 2004; Magid & Colder, 2007; Miller et al., 2003). Lack of Perseverance became significantly related to Disinhibition once Reward and Punishment sensitivity were included in the model, although it is presently unclear why this unexpected result occurred. Research using the Barratt Impulsiveness Scale 11 (BIS-11; Stanford et al., 2009) suggests that difficulties with executive inhibition may contribute to the relationship between lack of Premeditation and Disinhibition. Whiteside and Lynam (2001) created the Premeditation scale to measure a dimension underlying existing impulsivity scales including the Nonplanning and Motor Impulsiveness subscales of the BIS-11. The BIS-11 has so far been more widely used in neuropsychological research than the UPPS-P, and may provide further insight into the cognitive processes underlying lack of Premeditation. Although the results are mixed, it appears that the BIS-11 scales are related inversely to response inhibition (e.g., Keilp, Sackeim, & Mann, 2005). It may be that lack of Premeditation is influenced by reduced activity in prefrontal cortical regions involved in inhibitory control (see Congdon & Canli, 2008 for a review). Poor executive inhibition appears to share genetic influences with Externalizing (Young et al., 2009) and Premeditation may partially mediate this source of risk. Lack of Perseverance may also be related to poor attention leading to executive control problems.

Reward and Punishment Sensitivity were associated with Disinhibition; however, Punishment Sensitivity did not have a significant partial relationship with this component. This result is consistent with past findings of Reward Sensitivity being more reliably linked with Externalizing than is low Punishment Sensitivity (e.g., Hundt et al., 2008). Gullo, Ward, Dawe, Powell, and Jackson (2011) argued that Reward Sensitivity is an impulsivity facet reflecting heightened tonic activity in the mesolimbic dopamine system of the BAS and is separate from traits related to poor executive inhibition subsequent to relatively reduced prefrontal cortical activity, including perhaps Premeditation and Perseverance. It may be that the role of the BIS in influencing Disinhibition and Punishment Sensitivity is only significant when levels of BAS activation are high in the presence of possible punishment, and therefore the relationship between trait Punishment Sensitivity and Disinhibition may be contingent on trait levels of Reward Sensitivity.

Historically, Punishment and Reward Sensitivity have been viewed as having additive effects on motivating behavior, and this appears to be the case for individuals who are extremely high on the traits, however, the Joint Subsystems Hypothesis (Corr, 2004) within the RST suggests that under some circumstances stimuli activating the BAS will inhibit FFFS/BIS activity. As such, the lack of a unique relationship between Disinhibition and Punishment Sensitivity may be due to Reward Sensitivity not only responding to appetitive stimuli, but also inhibiting responsiveness to aversive stimuli.

While there is conceptual overlap between Sensation Seeking and Reward Sensitivity both remained associated with unique variance in Disinhibition when considered together in the regression. Sensation Seeking has been described as having two aspects. The first is the tendency to seek out and enjoy exciting experiences, while the second is openness to trying new potentially dangerous experiences (Whiteside & Lynam, 2001). The former may partially overlap with Reward Sensitivity while the latter may allow Sensation Seeking to act as a unique predictor. This latter aspect may overlap with insensitivity to punishment, thus, providing an alternate explanation of the loss of Punishment Sensitivity as a predictor when partial relationships were examined. This proposal is supported by our finding that of the UPPS-P scales Sensation Seeking had the greatest overlap in variance with Punishment Sensitivity.

4.2. Aggression

Negative Urgency was associated with Aggression, consistent with past findings (Derefinko, DeWall, Metze1, Walsh, & Lynam, 2011; Lynam & Miller, 2004; Miller et al., 2003; Settles et al., 2012). Those high on this trait may lash out at others when experiencing dysphoric mood. Similar to past findings, lack of Perseverance was not significantly associated with Aggression when the UPPS-P scales were considered alone. However, it was significantly related to Aggression, and Sensation Seeking was no longer, when the variance attributed to Reward and Punishment Sensitivity was taken into account. This suggests that some aspect of Perseverance that is not shared with Reward or Punishment Sensitivity may overlap with variance in Aggression.

Poor attention capacity may increase the impact of salient stimuli at the expense of other aspects in an interpersonal encounter and this may lead to aggressive behavior when attention is focused predominately on the more frustrating or provocative aspects of a situation. Giancola, Duke, and Ritz (2011) demonstrated experimentally that alcohol intoxication increases the impact of aggression relevant cues, presumably because alcohol reduces attentional capacity. A similar phenomenon, independent of Reward and Punishment Sensitivity, may occur in everyday life for people low on Perseverance. This trait may be associated with chronic limitations on attention capacity akin to those that may be acutely induced by alcohol. This, in turn, may lead to a habitually increased focus on conflict-laden aspects of interpersonal experiences.

High Reward Sensitivity and low Punishment Sensitivity were associated with Aggression even when the UPPS-P scales were taken into account. This is consistent with past findings of a positive relationship between Reward Sensitivity and Aggression (Bjørnebekk, 2007). An individual high on Reward Sensitivity may experience greater negative affect when they fail to obtain an anticipated reward in interpersonal situations leading to aggressive behavior. Furthermore, low Punishment Sensitivity may put individuals at greater risk for aggressive responses because they fail to make use of cues signaling impending punishment that might otherwise influence them to consider a different course of action. The presence of partial relationships for Aggression and both Reward and Punishment Sensitivity, suggests that Punishment Sensitivity is not related to Aggression exclusively because of the antagonistic effect of high Reward Sensitivity on BIS/FFFS activity.

4.3. Limitations

The present study had several limitations. Given the cross-sectional design, directional conclusions cannot be drawn. Replication with a prospective design is needed. Furthermore, our measures were limited to face valid self-reports. To reduce self-report bias, it may be beneficial to use a multi-method assessment involving collateral informants or behavioral measures purported to tap BIS and BAS activity. Our measures of Externalizing focused on overt behaviors commonly assessed in adults (Krueger et al., 2007), but our results may have differed if we also measured aspects common in child research on Externalizing such as inattention or hyperactivity (Hundt et al., 2008; Young et al., 2009). The generalizability of the results may be limited given that the sample was comprised of undergraduates, and although Externalizing appears to be a dimension extending throughout the population (Krueger et al., 2007), our results should be replicated in both clinical and population-representative samples.

5. Conclusion

Impulsivity is a multi-faceted construct with some aspects more strongly related to Externalizing behavior than others. The UPPS-P Impulsive Behavior scale provides an opportunity to measure several facets; however, it does not completely cover relevant aspects of impulsivity derived from the RST. As a result, in order to better understand Externalizing behavior, the UPPS-P should be used in conjunction with measures of Reward and Punishment Sensitivity.

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